

Economic Growth and Share of Temporary Workers

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Abstract

A dual labor market structure that consists of “permanent jobs” and “temporary jobs” is common in many Continental European countries and in Japan, and over the last two decades, the share of temporary workers in these countries has increased markedly. In this paper, I first demonstrate through an analysis of Japanese household panel survey data the difference in the wage structure and labor market conditions between permanent workers and temporary workers. Then, building a search and matching model of dual labor market with endogenous human capital accumulation, I show that, in the presence of two different types of jobs with different rates of wage growth, a slowing of the economic growth rate in a dual labor market structure can prompt a substantial shift in the composition of jobs toward temporary jobs.

Keywords Dual labor market, Growth, Unemployment, Temporary employment.

JEL classification E24, J64, O40

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1 Introduction

This paper studies the effect of the rate of economic growth on the composition of the labor market when that market consists of two types of employment, which are referred to as “permanent employment” and “temporary employment.”¹ In many OECD countries including Continental European countries and Japan, the share of temporary workers among all employees has increased markedly over the last two decades. This increasing trend has posed a serious concern for wide public because temporary workers, on average, earn less wages, expect very low rate of wage growth, receive less generous benefits, and experience weaker job security than permanent workers do.

Among the countries that have experienced a sharp rise in share of temporary workers, this paper focuses on Japan because: (i) the country experienced both periods of fast growth before 1990 and slow growth after 1990, and during the slow growth period, the share of temporary employment has nearly doubled from 20% to 40%; (ii) the presence of two distinctively different types of jobs is well documented²; and (iii) panel household survey was readily available, which enabled me to study different characteristics of permanent and temporary workers. Figure 1 shows two key Japanese economic statistics between 1980 and 2015. First, Japan experienced periods of both high growth prior to 1990 and low growth post-1990. The slow growth period after 1990 is sometimes referred to as the lost decades. The difference between the two periods is evident in the first panel (1a). The dotted lines indicate the average Japanese real GDP growth prior to and post 1990, which are 4.47% and 0.90%, respectively.

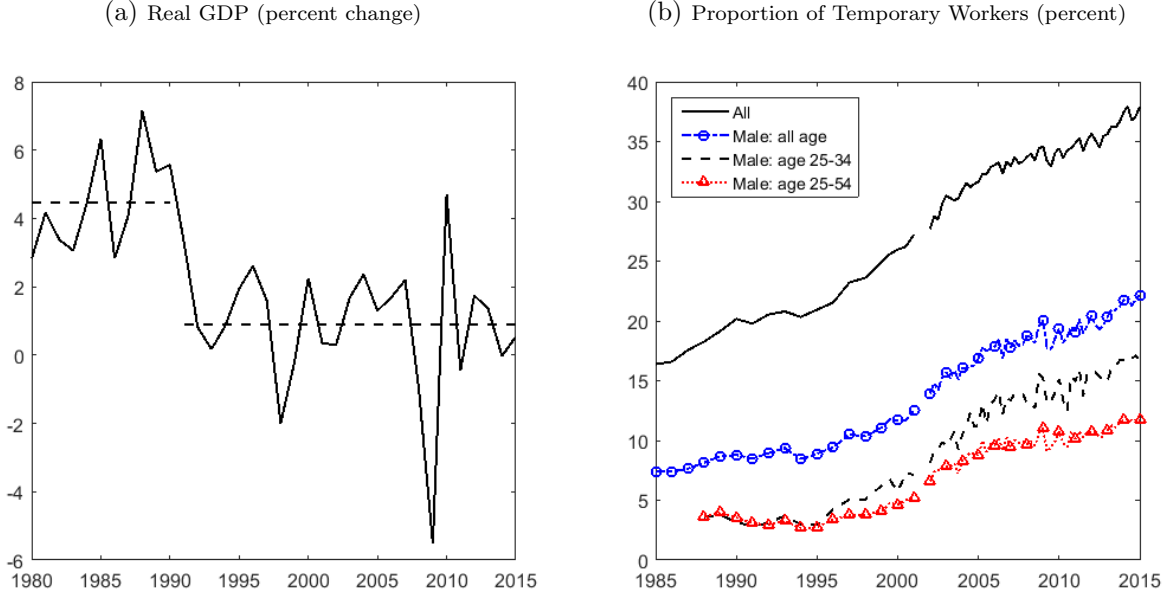
During the period of slow growth, the share of temporary workers among all employees doubled from 20% to close to 40%, as illustrated in panel 1b.³ The share of temporary workers increases even when the sample is restricted to male employees and male employees of working age. This outcome implies that the increasing trend is likely to persist even after factors such as greater participation of female workers and aging demographics are removed.

¹How these types of employment are referenced in the literature varies. Examples include “good jobs” versus “bad jobs”, “standard jobs” versus “non-standard jobs”, “full-time work” versus, “part-time”, and “indefinite employment” versus “fixed-term employment”.

²Precise definitions of “permanent jobs” and “temporary jobs” are given in Section 3.

³Data are retrieved from the Labor Force Survey, whose survey design is similar to that of the Current Population Survey in the U.S. Survey was redesigned in 2001, but an overall increasing trend in the share of temporary workers is evident across the entire population.

Figure 1: Japanese Economy



Source for 1a: SNA (National Cabinet of Japan)
Source for 1b: "The Special Survey of the Labour Force Survey" from 1984 to 2001,
"Labour Force Survey (Detailed Tabulation)" since 2002

Notably, the most drastic change is observed for male workers between the ages of 25 and 34. This is likely due to the Japanese employment practice where majority of permanent employment contracts are formed when workers are in their 20s and 30s. Thus, the dramatic increase in the share of temporary workers for these young men, in contrast to old men whose temporary employment rate increased only modestly, suggests that the job-finding margin plays more important role in explaining the increasing share of temporary workers than the job-separation margin do.

In a large literature that studies the increasing share of temporary workers, the trend is commonly attributed to changes in labor market regulations that enforce stronger job protection or legislative decisions that facilitate the creation of temporary jobs.⁴ While these legislative factors may play some role, this paper argues that, under a certain condition, a slowdown of economic growth alone can create a considerable increase in the share of temporary workers even in the absence of any legislative change.

To formalize our intuition, I build a search and matching model of a dual labor market

⁴See Cahuc et al. (2016) and Boeri et al. (2011).

that is populated by heterogeneous workers whose skills evolve endogenously according to their employment status. The model incorporates a dual labor market that is designed to capture two distinctive labor markets for permanent jobs and temporary jobs. Workers are born as initially unemployed and with randomly assigned skills. Depending on their skill levels, they decide to search either in the permanent labor market or in the temporary labor market. In choosing which market to search for, unemployed workers face the following trade-off. If the unemployed workers search in the permanent market, they have a chance of finding jobs that enable them to fully utilize their own skills and that ensure greater job-security and on-the-job training. However, the probability of finding such jobs is low. In contrast, if the unemployed workers search in the temporary market, they have higher probability of finding a job. Yet, workers holding these jobs face weaker job-security and receive less on-the-job training. In addition, the workers holding these jobs cannot utilize their own skills; instead they employ a technology that is common to all temporary jobs. Given the parameter values that I estimate in Section 6, low skilled workers find it optimal to search for temporary jobs while high skilled workers search for permanent jobs; the observation that is consistent with the data.⁵

Given this trade-off between permanent jobs and temporary jobs, I pose the following question: How does low growth affect labor market composition? Suppose that an economy enters a slow growth steady state. Slower growth means that future income is discounted at a higher rate. This reduces the continuation value of a job because the cost of setting up the initial employment relationship is paid now; in contrast, the benefit of the employment relationship accrues in the future.⁶ When the workers' skills are allowed to increase on the job and wages are negotiated each period, the value of permanent jobs decreases at a much faster rate than the rate at which the value of temporary jobs decreases. This is because when the growth rate of aggregate economy goes down, future wage growth is discounted at a higher rate, making it less valuable to hold jobs that promise future wage growth (permanent jobs). Faced with the decreasing benefit of permanent jobs, some workers who would search for permanent jobs in a fast growth economy will search for temporary jobs in a slow growth economy.

At the same time, firms reduce vacancy postings of permanent jobs at a greater margin

⁵See Section 3.2.

⁶This effect is often called the “capitalization effect,” which is highlighted by [Aghion and Howitt \(1994\)](#) and [Pissarides \(2000\)](#).

than they do for vacancy postings of temporary jobs due to the relatively larger reduction of the value of the temporary jobs, which makes the search for permanent employment further unattractive for the unemployed. Consequently, the share of permanent workers decreases in the slow growth economy. The extent to which the share of permanent workers decreases depends on the rate of skill accumulation for the permanent workers. In fact, when human capital accumulation is muted, slower economic growth has no impact on the composition of the labor market. When estimated with the Japanese labor market dataset, my model predicts that about 70% of the observed increase in the share of temporary workers among Japanese male can be explained by its slow growth.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 provides background information about the Japanese labor market and it defines “permanent jobs” and “temporary jobs” in the context of the Japanese labor market. Section 4 illustrates the mechanism using a simple model. Section 5 describes the full model. Section 6 lays out the calibration strategy, Section 7 shows the result, and Section 8 concludes.

2 Related Literature

This paper contributes to the literature on growth and unemployment. A number of papers have studied the effect of growth on unemployment in a search and matching framework (Aghion and Howitt (1994), Pissarides (2000), Pissarides and Vallanti (2007)). One of the most heated debates has centered on the degree to which the new workers embody new technology. If all the technological progress is embodied by new workers, then growth and unemployment are positively correlated. In contrast, if technological growth is disembodied, meaning that both new and existing workers embody technological progress, then growth and unemployment will be negatively correlated. To address this issue, Pissarides and Vallanti (2007) evaluate a model that features embodied and disembodied technology, capitalization, and creative destruction effects. They conclude that embodied technology and creative destruction do not play a significant role in explaining the steady-state unemployment rate. Following their argument, this paper uses a search and matching model with disembodied technological progress.

The use of temporary workers in Japan has gained attention in recent years.⁷ [Asano, Ito and Kawaguchi \(2013\)](#) empirically examined factors that are most responsible for the rise of temporary employment in Japan. They found that changes in labor force and industrial composition account for only a quarter of the increase in temporary workers. In section 9.6, I explain that the timing of legislative changes does not align well with the timing of the increase in share of temporary workers. In addition to these factors such as industrial composition, changes in labor force, and legislative changes, the current paper suggests that the decline of aggregate economic growth rate together with the difference in the rates of human capital accumulation between permanent and temporary jobs as the major source of the observed rise of the temporary workers share in Japan.

The paper builds on the framework of [Huckfeldt \(2016\)](#), who proposes a search model where the allocation of workers across different labor markets is endogenously determined. He then studies how this allocation changes over the business cycle. In contrast, I study how the allocation of workers across distinct jobs changes across growth regimes. In doing so, I identify a separate mechanism by which the impact of changes in growth rates on the allocation of workers depends on the difference in the rates of wage growth across permanent and temporary jobs. This paper is also related to research undertaken by [Acemoglu \(2001\)](#), who developed a search and matching model in which high-wage (“good jobs”) and low-wage (“bad jobs”) jobs can coexist. Observing the increasing share of temporary employment in Continental European countries, [Alonso-Borrego et al. \(2005\)](#) investigated the role of increased temporary employment in a general equilibrium search and matching model. He concluded that temporary jobs increase unemployment, reduce output, and raise productivity. [Caggese and Cuñat \(2008\)](#) studied the relationship between financial constraints and the labor market composition of permanent and temporary jobs. They found that when firms are likely to be financially constrained, they choose to hire more temporary workers. Examining the experience of labor market reform in France, [Blanchard and Landier \(2002\)](#) discuss the effect of allowing firms to hire workers on fixed-term contracts. They conclude that the main effect of policy reform is a high turnover in entry-level jobs, which leads to higher unemployment.

In the area of the growth and the composition of jobs in the economy, this paper is most

⁷For example, in a structural estimation of the career choices of young workers, [Esteban-Pretel, Nakajima and Tanaka \(2011\)](#) focus on young workers who began their careers as temporary workers. They conclude that in Japan starting a career as a temporary worker has a lasting effect on the welfare of young workers.

closely related to [Miyamoto \(2016\)](#) and [Wasmer \(1999\)](#). In a search and matching framework, [Wasmer \(1999\)](#) claims that through the capitalization effect, the share of temporary workers increases during times of low technological growth. [Miyamoto \(2016\)](#) builds a search and matching model to quantify the strength of the capitalization effect in Japan. Although our papers are similar, his mechanism crucially depends on the substitutability between goods produced by permanent workers and goods produced by temporary workers. As the rate of substitution between permanent goods and temporary goods decreases, the effects of a change in trend growth on the composition of jobs also decreases. Given how difficult it is to measure the rate of substitution between permanent goods and temporary goods in the real world, I propose an alternative mechanism that relies on the difference in the rate of wage growth between the two types of employment by integrating heterogeneous workers' skills that evolve dynamically. Taking skill accumulation into account, I find that the effect of the rate of growth on labor market composition changes dramatically.

3 “Permanent Job” and “Temporary Job” in the Japanese Labor Market

This section outlines the institutional background of the Japanese labor market, whose composition is the focus of this paper. In particular, I describe standard Japanese labor market practices that many believe differ from US practices, and I define “permanent employment” and “temporary employment” in the context of Japanese labor market practices. [Kambayashi and Kato \(2012\)](#) summarize as follows a set of standard practices that are coherently observed in Japanese labor market:

1. Strong employment protection against lay-offs, which leads to a long job tenure.
2. Employee involvement in problem solving activities from the bottom-up, which encourages workers to exert discretionary effort and gain local knowledge.
3. Careful screening and extensive on-the-job training that increases worker ability⁸

⁸[Kambayashi and Kato \(2012\)](#) also highlighted two other Japanese employment practices: (4) A robust scheme to share information through unions so that information asymmetry between employees and between employees and management is avoided; and (5) Incentive schemes that include employee ownership of the firms and profit sharing, which encourage workers and the firm to develop a sense of shared interests.

The workers who are covered by these standard practices I call “permanent workers.” Another group of Japanese workers has an employment relationship that does not conform to these practices. They constitute the secondary segment of the Japanese labor market, and I refer to their situation as “temporary employment.” These workers often are paid lower wages, receive less generous benefits, enjoy less control over their work, and have weaker employment protection than permanent workers ([Kambayashi and Kato \(2012\)](#)). In this regard, the permanent and temporary jobs of the Japanese labor market are similar to the “good jobs” and “bad jobs” that are described in the good job and bad job literature and “permanent jobs” and “temporary jobs” in the literature that study European labor markets ⁹.

In the following sub-sections, I document three empirical facts about permanent and temporary workers: (i) the rate of wage growth is higher for permanent workers and, on average, the job tenure of permanent workers is greater than that of temporary workers, (ii) people with high school or middle school degrees are more likely to become temporary workers while people with college or higher degrees are more likely to become permanent workers, and (iii) the opportunity to receive training on the job for temporary workers is about half of that for permanent workers.

The following empirical findings are based largely on two datasets: the Japanese Household Panel Survey (JHPS) and the Keio Household Panel Survey (KHPS). The KHPS is an annual panel household survey of 4000 households that was recorded each year from 2004 to 2013. The JHPS is another annual panel household survey of about 4000 households that was recorded each year from 2009 to 2013. I focus on the income of household heads aged 25 to 60 years who were recorded at least 4 times (waves). This criterion leaves us with about 2000 observations each year from 2004 to 2008 and about 3200 observations from 2009 to 2013. The classification of permanent worker and temporary worker follows the criterion described above.

In these datasets, I define “temporary employees” as workers who fall into one of the following categories: (i) part-time worker, (ii) dispatched workers, (iii) contract workers, (iv) commission staff, or (v) other types of workers who are not permanent employees. This method of classification is consistent with the use of the term in the Labor Force Survey

⁹See, for example, [Acemoglu \(2001\)](#) and [Kalleberg \(2011\)](#)

¹⁰ and capture the different employment practices for permanent and temporary workers described above. Under this system of classification some temporary workers work for more than 40 hours a week and some permanent workers can work less than 40 hours a week.

3.1 Wage growth and job tenure

First, I establish that there is a difference in the rate of wage growth between permanent workers and temporary workers. To identify the return on working at the same firm for additional year, I run a regression of log hourly wage on job tenure and job tenure square and their interaction terms with the indicator for permanent workers. Additionally, the regression includes the following control variables in $\mathbf{X}_{i,t}$: age, age squared, and fixed effects for year, industry, and region. Let I_P be the indicator for permanent employment, then the regression can be specified as follows:

$$\begin{aligned} \text{wage}_{i,t} = & \alpha_i + \alpha_1 \text{tenure}_{i,t} + \alpha_2 \text{tenure}_{i,t}^2 \\ & + \beta_0 \times I_{P,i,t} + \beta_1 \text{tenure}_{i,t} \times I_{P,i,t} + \beta_2 \text{tenure}_{i,t}^2 \times I_{P,i,t} + \mathbf{X}_{i,t} \Phi + \epsilon_{i,t}, \end{aligned}$$

where α_i represents individual fixed effects. Our parameters of interest are β_1 and β_2 . If the rate of wage growth is different across two types of employment, the β 's should be jointly significant. The result is shown below in Table 1. First, the p -value of the F -test indicates that the null of joint insignificance of β 's is rejected at 5% significance level, meaning that the annual rate of return for working at the same firm is different between permanent workers and temporary workers. For example, the average return of staying at the current firm for permanent workers with 3 years of job tenure (median years of job tenure for temporary workers), even after removing the effect of aging, is 1%. In contrast, the annual return of working at the current firm is essentially zero for temporary workers with the same years of job tenure. The difference in the rates of wage growth in this case is about 0.978%. The difference in the rate of wage growth stays around 1% even when we consider a worker with job tenure of 11 years. The result is consistent with common practices documented in Section 3 wherein permanent workers accumulate local knowledge within a firm and receive extensive on-the-job training; temporary workers, in contract, do not.

¹⁰The survey design is similar to the U.S. Current Population Survey

Table 1: Return on working

	Permanent	Temporary
Median years of job tenure	11 yrs	3 yrs
Annual wage growth for workers with 3 yrs of tenure	1.005%	0.027%
Difference in wage growth for workers with 3 yrs of tenure	0.978%	
Annual wage growth for avg. workers with 11 yrs of tenure	0.946%	-0.02%
Difference in wage growth for avg. workers with 11 yrs of tenure	0.971%	
Observations	14,303	
R^2	0.1018	
p -value for F -test ($\beta_1 = \beta_2 = 0$)	0.0337	

Data are from KHPS and JHPS, 2004-2013. For each year, the sample is restricted to workers who continued working at the same firm from the previous year.

Table 1 also shows that the median years of job tenure for permanent workers is 11 years while the median years of job tenure is 3 years for temporary workers. For comparison, the median years of staying with current employers for American workers is 4.2 years on January 2016 according to the Bureau of Labor Statistics. Not surprisingly, the average years of job tenure is substantially higher for permanent workers than it is for temporary workers. The data shows that the average length of stay with the same employer is 13.308 years (with standard deviation of 10.414) for permanent workers and is 4.637 years (with standard deviation of 5.567) for temporary workers. The test for the difference in the mean rejects the null that across the two types of workers the means are equal at the 1% significance level.

3.2 Educational attainment

The assumptions in my model induce low skilled workers to search for temporary jobs and high-skilled workers to search for permanent jobs; I present an observation that is consistent with this prediction. Since workers' innate productive capacities are difficult to measure, I use workers' educational attainment as a proxy for their skills. The share of temporary workers is 26% in 2012 among male employees with middle or high school degrees. In contrast, the share of temporary workers is 12% in 2012 among male employees who have college or higher

degrees, which is less than half of the shares among college graduates. While this is not a perfect measure of workers' skills, it hints at the difference in the workers' traits across the two types of employment as measured by educational achievement.

3.3 On the job training

Lastly, I document that firms are about twice more likely to provide job-training for permanent workers than they do for temporary workers. Figure 2 shows a percentage of establishments who provided job-training for their employees. The black solid bar indicates the percentage of establishments who provided job-training to their permanent employees, and the shaded bar shows the percentage of establishments who provided job-training for temporary employees.¹¹ The figure shows that about 60% of establishments provides “on-the-job training” to their permanent employees within a given year while only about 25% of them provides temporary employees with the “on-the-job training.”¹² Additionally, approximately 70% of the establishments provides permanent workers with “off-the-job training” while only about 32% of them provides temporary workers with the training. Figure 2 illustrates the degree to which temporary workers lack opportunity to improve their skills while they are on the job.

Having documented these facts, I now present my model in the following section.

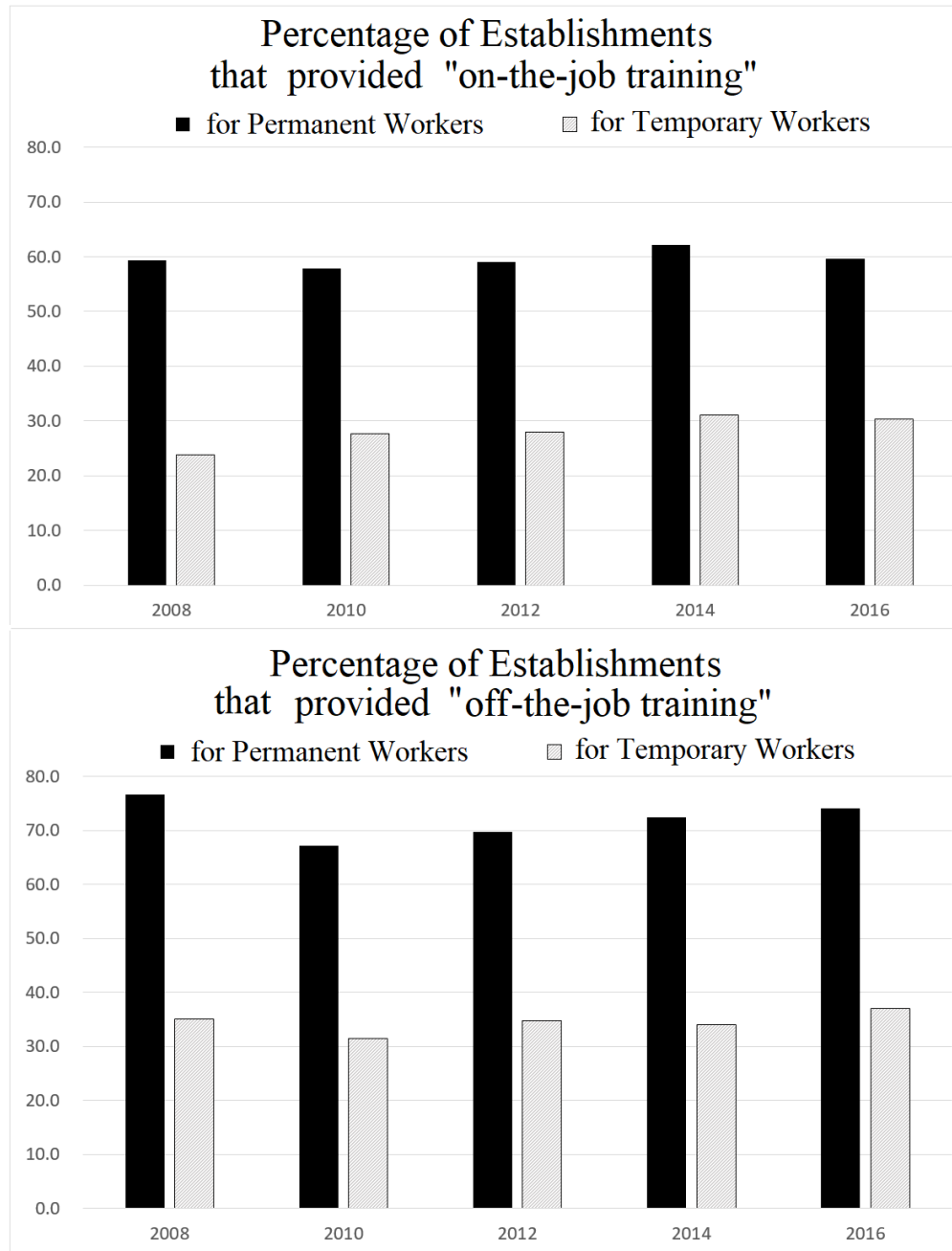
4 Simple Model

For an illustration of a mechanism, I first build a simple model that features two types jobs. For convenience, I refer to the first type of job a permanent job and the second type of the job a temporary job. These two types of jobs are identical except that wages for permanent workers exhibit growth while wages for temporary workers remain fixed for their entire career. Workers employed in the permanent jobs initially receive wages, w_{perm} . In the next period, their wages increase to w'_{perm} , which will then be fixed for the rest of their

¹¹ “on-the-job training” is defined as a form of training that is given while employees are conducting regular task but is heavily supervised. This type of training is usually well-planned beforehand. “off-the-job training” refers to a form of training that is given while employees are not engaging regular tasks.

¹² Some may argue that this figure simply reflects the difference in the number of permanent and temporary employees in each establishments. I am currently estimating a probability of receiving a training for permanent and temporary workers from the household panel survey.

Figure 2: “on-the-job” and “off-the-job” training



Source: Basic Survey of Human Resources Development

careers. I further assume that once attached, the employed workers do not separate from the jobs.

Besides the wage growth that is specific to the permanent workers, the wages of both

sectors will increase at the common rate of γ_X , which reflects the growth rate of the economy.

Let W_{perm} denote the present value of a permanent worker. Then,

$$W_{\text{perm}} = w_{\text{perm}} + \beta\gamma_X \frac{w'_{\text{perm}}}{1 - \beta\gamma_X}, \quad (1)$$

where β is a discount factor.

Temporary workers do not experience wage growth. Thus, the value of a temporary worker is:

$$W_{\text{temp}} = \frac{w_{\text{temp}}}{1 - \beta\gamma_X} \quad (2)$$

An unemployed worker is indifferent between searching for a permanent job and a temporary job if the following condition holds:

$$W_{\text{perm}} = W_{\text{temp}}$$

This condition requires the following wage structure: $w'_{\text{perm}} > w_{\text{temp}} > w_{\text{perm}}$

Suppose now that the technological growth that is common to both types of workers decreases, $\gamma_X \downarrow$. Then, values of both permanent workers and temporary workers decrease. For the unemployed workers to continue to be indifferent between choosing between permanent jobs and temporary jobs, the values of a permanent worker and that of a temporary worker must both go down at the same rate, $dW_{\text{perm}}/d\gamma_X = dW_{\text{temp}}/d\gamma_X$. On the contrary, the unemployed workers prefer temporary jobs if the value of permanent workers decreases at a greater rate than the rate at which the value of temporary workers decreases: $dW_{\text{perm}}/d\gamma_X > dW_{\text{temp}}/d\gamma_X$.

It turns out that if the permanent jobs feature wage growth while the temporary jobs do not, then the rate of reduction in the value of permanent jobs is larger in response to the decrease in γ_X as follows:

$$\frac{dW_{\text{perm}}}{d\gamma_X} = \frac{\beta w'_{\text{perm}}}{(1 - \beta\gamma_X)^2} > \frac{\beta w_{\text{temp}}}{(1 - \beta\gamma_X)^2} = \frac{dW_{\text{temp}}}{d\gamma_X}$$

since $w'_{\text{perm}} > w_{\text{temp}}$.

Because changes in the discount factor affects only through wages in the future periods, the rate of response is larger for the permanent sector where future wage is higher than the temporary wages. Thus, the unemployed workers who were previously indifferent between permanent jobs and temporary jobs now prefer to be employed in temporary jobs when γ_X decreases. This simple model, however, does not take into account the general equilibrium effects of a slow growth. Specifically, when more people search for temporary jobs, the temporary labor market becomes tighter, which makes finding a temporary job more difficult and searching in the temporary labor market less attractive (“congestion effects”). At the same time, this model also does not take into account the firms’ reaction to the economic slowdown. Importantly, the simple model also ignore the effects via the wages. To fully evaluate the extent to which a slow growth affect the labor composition, I build a full model in the following sections.

5 Full Model

5.1 Environment

The model builds on [Huckfeldt \(2016\)](#) who incorporates elements of the Diamond-Mortensen-Pissarides search and matching model and the [Ljungqvist and Sargent \(1998\)](#) model of human capital accumulation and depreciation

The economy is populated by risk-neutral workers and firms. Each employment relationship consists of a worker-firm pair. Workers are born unemployed and have skills that are drawn from a log-normal distribution: $\log(h) \sim N(0, \sigma_h^2)$. The skills of each worker dynamically evolve according to his or her employment status, as specified below.

Two labor markets comprise my model. The first is the permanent job market and the second is the temporary job market. These two employment types differ from one another in two respects: (i) while a firm must pay a firing cost to dissolve a permanent match, there is no cost of dissolving a temporary match; and (ii) permanent workers use their skills in the production process, in contrast to temporary workers, who do not utilize their skills. I assume that firing costs constitute waste rather than transfer to workers (i.e., administrative and legal costs).

Once an employment relationship is formed, firms produce goods according to the following linear technologies:

$$\tilde{y}_{\text{perm}} = \underbrace{X}_{\substack{\text{labor-augmenting} \\ \text{technology}}} \times \underbrace{h}_{\text{skill level}}, \quad \tilde{y}_{\text{temp}} = \underbrace{X}_{\substack{\text{labor-augmenting} \\ \text{technology}}} \times \underbrace{\bar{A}}_{\substack{\text{temporary} \\ \text{technology}}}$$

I assume that while permanent workers can make use of their skills, temporary workers use a technology that is commonly available to all temporary workers. The productivity of temporary workers is fixed at $\bar{A}$.¹³ To analyze the effect of growth, I assume that there is a labor-augmenting technology, X .¹⁴ Additionally, I assume that all separations occur for exogenous reasons. The assumption of exogenous separation is not necessary for the effect of slow growth on job composition to be propagated.

That there is no on-the-job search reflects the fact that in Japan a direct transition from temporary employment to permanent employment is rare. [Esteban-Pretel et al. \(2011\)](#), who conducted a structural estimation of the career choices of young workers in Japan, found that temporary employment is seldom a stepping stone to permanent employment. My assumption that there is no job-to-job transition is consistent with the observed extremely long job tenure documented in Section 3.1. Finally, I focus on stationary stochastic equilibrium.

5.2 Labor Markets Construction

There are two labor markets. During each period, unemployed workers with skill level, h , choose in which market to search for a job. The search is random within the same market.

The match is formed according to a Cobb-Douglas function that takes job-seekers and vacancies as factors of input. Let M denote the number of matches and let u and v be the number of job-seekers and vacancies, respectively. The match is formed according to the following:

$$M_j = \bar{m}_j u_j v_j^{1-\psi} \quad \text{where } j \in \{\text{perm}, \text{temp}\}$$

, where $\bar{m}_j$ is a constant match efficiency for j labor market. The match efficiency is different across two markets. For convenience, I define market tightness as the ratio of vacancies per

¹³This productivity structure for temporary workers implies that temporary jobs are “skill-neutral”.

¹⁴See Section 5.3 and 9.2 for more detail

job-seeker: $\theta_j = v_j/u_j$. Then, the job-finding probability can be defined as $f(\theta_j) = \bar{m}_j \theta_j^{1-\psi}$ and the job-filling probability as $q(\theta_j) = \bar{m}_j \theta_j^{-\psi}$ for $j \in \{\text{perm}, \text{temp}\}$.

The presence of the firing cost changes the value functions of the permanent workers and firms as well as the wage structure of the permanent workers. Following [Mortensen and Pissarides \(1999\)](#) and [Pries and Rogerson \(2005\)](#), I assume that firms that reject new permanent workers do not have to pay firing costs because their employment relationships have not yet started. Once they start working, the firms must pay firing costs when their matches are destroyed. This setting gives rise to a two-tier wage structure for permanent jobs, which is often described to as an “inside” wage and “outside” wage ([Lindbeck et al. \(1989\)](#)). I denote w_{perm}^N to be the wage for a newly employed permanent worker and w_{perm}^E to be the wage of an existing permanent worker. Because of the presence of the two-tier wage structure, there have to be two value functions for the permanent workers and jobs.

5.3 Balanced Growth

Following [Pissarides \(2000\)](#) and [Pissarides and Vallanti \(2007\)](#), my model assumes disembodied technological growth. I assume that there is a labor-augmenting technology, X_t that exhibits a steady state rate of growth as $\gamma_X = X_{t+1}/X_t$ for all t . Furthermore, I assume that all exogenous variables grow at the constant rate of γ_X . Thus, the cost of vacancy, κ_j , the unemployment income, b , and the firing cost, ϕ , for $j \in \{\text{perm}, \text{temp}\}$ all grow at the rate of γ_X . Under this assumption, the economy is on a balanced growth path. In the following sections, I present a model under which all variables are divided by X (i.e. $y_{\text{perm}} = \tilde{y}_{\text{perm}}/X$). For a detailed derivation, please refer to [Section 9.2](#).

5.4 Human skill dynamics

The workers in my model are endowed with skills. The set of skills are measured between $\underline{h}$ and $\bar{h}$, and each skill-level is located at Δ_h distance from the others. Human capital in my model evolves dynamically. Given that wages are negotiated each period, the wages grow as workers’ skills upgrade while on-the-job. This is one way to capture the wage growth described in [3.1](#). According to [Rubinstein and Weiss \(2006\)](#), in the U.S. there are two main sources for an observed earnings/experience profile: (1) skill accumulation and (2) job

search (including outside job offers). In my model, workers increase their wages only by accumulating more skills. My model ignores the wage growth from an on-the-job search because in Japan the job-to-job transition rate is small. As noted in Section 3.1, the rate of wage growth is different across the two types of employment. To capture the different rate of wage growth, I denote π_{perm} as the probability of a skill upgrade by Δ_H for permanent workers and π_{temp} as the probability of a skill upgrade for temporary workers. Thus, during each period, π_j fraction of workers in sector j who have an h skill-level upgrade their skills to $h + \Delta_h$ in the next period. In this model, however, even though temporary workers can upgrade their skills, they cannot utilize their skills at their jobs.

In addition, unemployed workers face the probability of skill deterioration with the probability, π_u . Thus, for unemployed workers who have skill-level, h , their skills go down to $h - \Delta_h$ each period with the probability π_u .

5.5 Value of Unemployment

Let $U(h)$ denote the present value of an unemployed worker who has h skill-level.

$$U(h) = b + \beta(1 - \nu)\gamma_X \max \mathbb{E} [U_{\text{perm}}(h'), U_{\text{temp}}(h')] \quad (3)$$

Unemployed workers first receive an unemployment income, b , and decide whether to search in the permanent market or the temporary market.

Let $U_{\text{perm}}(h)$ denote the present value of unemployed workers who seek employment in the permanent labor market, and let define $U_{\text{temp}}(h)$ as the present value of unemployed workers who search in the temporary market. Then,

$$U_{\text{perm}}(h) = f(\theta_{\text{perm}})(W_{\text{perm}}^N(h) - U(h)) + U(h), \quad (4)$$

$$U_{\text{temp}}(h) = f(\theta_{\text{temp}})(W_{\text{temp}}(h) - U(h)) + U(h), \quad (5)$$

where $\beta = 1/R$ is the discount factor, ν is the exogenous probability of retirement, and γ_X is the steady state rate of labor-augmenting technological growth.

Once the unemployed worker decides what market to search, the unemployed worker finds a job with the probability $f(\theta_j)$ for $j \in \{\text{perm}, \text{temp}\}$. If the unemployed worker could

not find a job, then she will remain unemployed when the next period starts.

For reasonable parameter values, all workers with a skill level above a given cutoff h^* choose to search in the permanent market because the value of searching in the permanent market exceeds the value of searching in the temporary market. Thus, for $h > h^*$, $U_{\text{perm}}(h) > U_{\text{temp}}(h)$. Similarly, all workers below cutoff h^* find it optimal to search in the temporary market. An unemployed worker who held a temporary job previously might find it optimal to search for a permanent job if, when he worked in the temporary job, he accumulated sufficient skills. In contrast, an unemployed worker who previously held a permanent job will search in the temporary market if his skill drops to a level below h^* .

5.6 Value of Worker

Let $W_{\text{perm}}(h)$ denote the present value of a permanent worker. Let $W_{\text{perm}}^N(h)$ denote the present value of a new permanent worker:

$$W_{\text{perm}}^N(h) = w_{\text{perm}}^N(h) + \beta(1 - \nu)\gamma_X \mathbb{E} \left[(1 - \delta_{\text{perm}})(W_{\text{perm}}^E(h') - U(h')) + U(h') \right], \quad (6)$$

subject to the law of motion for h .

An existing permanent worker faces an exogenous probability of match destruction, δ_{perm} , at the beginning of each period. If the match is not destroyed, then the worker stays at the same firm. The worker then provides labor and receives wages. Let $W_{\text{perm}}^E(h)$ denote the present value of an existing permanent worker:

$$W_{\text{perm}}^E(h) = w_{\text{perm}}^E(h) + \beta(1 - \nu)\gamma_X \mathbb{E} \left[(1 - \delta_{\text{perm}})(W_{\text{perm}}^E(h') - U(h')) + U(h') \right], \quad (7)$$

subject to the law of motion for h . The only difference between the value of a new worker and an existing worker is the wage.

Finally, let $W_{\text{temp}}(h)$ denote the present value of a temporary worker.

$$W_{\text{temp}}(h, z) = w_{\text{temp}}(h) + \beta(1 - \nu)\gamma_X \mathbb{E} \left[(1 - \delta_{\text{temp}})(W_{\text{temp}}(h') - U(h')) + U(h') \right], \quad (8)$$

subject to the law of motion for h .

5.7 Value of a Job

Let $J_{\text{perm}}^N(h)$ denote the present value of a new permanent job

$$\begin{aligned} J_{\text{perm}}^N(h) = & y_{\text{perm}} - w_{\text{perm}}^N(h) \\ & + \beta(1 - \nu)\gamma_X \mathbb{E} \left[(1 - \delta_{\text{perm}})(J_{\text{perm}}^E(h') - V) + V - \delta_{\text{perm}}\phi \right], \end{aligned} \quad (9)$$

subject to the law of motion for h . The match continues to next period unless it is not exogenous with the probability δ_{perm} . If the match is destroyed at the beginning of next period, the firms pay the firing costs of ϕ . V is the value of posting a vacancy.

Let $J_{\text{perm}}^E(h)$ denote the present value of an existing permanent job

$$\begin{aligned} J_{\text{perm}}^E(h) = & y_{\text{perm}} - w_{\text{perm}}^E(h) \\ & + \beta(1 - \nu)\gamma_X \mathbb{E} \left[(1 - \delta_{\text{perm}})(J_{\text{perm}}^E(h') - V) + V - \delta_{\text{perm}}\phi \right], \end{aligned} \quad (10)$$

subject to the law of motion for h .

The value of an existing permanent job and the value of a new permanent job differ only in their wages. As explained above, wages differ because new permanent firms do not need to pay firing costs when they reject new workers.

Similarly, let $J_{\text{temp}}(h)$ define the present value of a temporary job:

$$\begin{aligned} J_{\text{temp}}(h) = & y_{\text{temp}} - w_{\text{temp}}(h) \\ & + \beta(1 - \nu)\gamma_X \mathbb{E} \left[(1 - \delta_{\text{temp}})(J_{\text{temp}}(h') - V) + V \right], \end{aligned} \quad (11)$$

subject to the law of motion for h . The value of temporary jobs is similar to that of permanent jobs except that temporary jobs do not incur firing costs.

5.8 Value of Vacancy and the Free Entry Condition

Finally, the value of posting a vacancy in the permanent market is

$$V_{\text{perm}} = -\kappa_{\text{perm}} + \beta(1 - \nu)\gamma_X \mathbb{E} \left[q(\theta_{\text{perm}}) \int_h (J_{\text{perm}}^N(h) - V_{\text{perm}}) d\mu_{\text{perm}}^h + V_{\text{perm}} \right], \quad (12)$$

where μ_{perm}^h is the skill distribution of unemployed workers in the permanent market. I assume that at the time of posting vacancies, the firms do not know the skill distribution of the job-seekers. The worker's skill is revealed only after he or she and the firm meet. Thus, the job-filling rate is the same across all skill-levels. The cost of posting a vacancy is defined as κ_{perm} for permanent jobs. The worker and firm meet at a job-filling probability given by $q(\theta_{\text{perm}})$. If the vacancy meets a worker, then a new job is formed whose continuation value is given by $J_{\text{perm}}^N(h, z)$.

The value of posting a vacancy in the temporary market is

$$V_{\text{temp}} = -\kappa_{\text{temp}} + \beta(1 - \nu)\gamma_X \mathbb{E} \left[q(\theta_{\text{temp}}) \int_h (J_{\text{temp}}(h) - V_{\text{temp}}) d\mu_{\text{temp}}^h + V_{\text{temp}} \right], \quad (13)$$

where μ_{temp}^h is the skill distribution of unemployed workers in the temporary market and κ_{temp} is the cost of posting a vacancy in the temporary market. Each vacancy meets the job-seeker with the probability of $q(\theta_{\text{temp}})$. In equilibrium, the free-entry condition drives the value of a vacancy in both markets down to zero: $V_{\text{perm}} = 0$ and $V_{\text{temp}} = 0$.

5.9 Wage Determination

Following [Pissarides \(2000\)](#), I assume that workers and firms negotiate wages such that they split the matching surplus according to the Nash Bargaining game. Following [Pries and Rogerson \(2005\)](#) and [Pissarides \(2000\)](#), I assume that there is a two-tier wage structure for permanent jobs. This assumption is motivated by the observation that newly employed permanent firms do not have to pay firing costs if they reject workers before the employment relationships begin. Once workers start working, firms have to pay their firing costs if they are laid off. Let η denote a worker's fixed bargaining power. According to the Nash Bargaining solution, the new permanent worker and the firm split the match surplus as follows:

$$(1 - \eta)(W_{\text{perm}}^N(h) - U_{\text{perm}}(h)) = \eta J_{\text{perm}}^N(h) \quad (14)$$

The existing permanent worker and a firm split the match surplus as follows:

$$(1 - \eta)(W_{\text{perm}}^E(h) - U_{\text{perm}}(h)) = \eta(J_{\text{perm}}^E(h) + \phi) \quad (15)$$

Finally, for the temporary worker and firm, the match surplus is divided as follows:

$$(1 - \eta)(W_{\text{temp}}(h) - U_{\text{temp}}(h)) = \eta J_{\text{temp}}(h) \quad (16)$$

Then, wages for new permanent workers can be shown as follows:¹⁵

$$\begin{aligned} w_{\text{perm}}^N = & (1 - \eta)b + \eta(y_{\text{perm}} - \beta(1 - \nu)\gamma_X(1 - \delta_{\text{perm}})\phi) \\ & + (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (\pi_{\text{perm}} - \pi_u)U(h) + \pi_u U(h - \Delta_h) - \pi_{\text{perm}} U(h + \Delta_h) \right\} \\ & + (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (1 - \pi_u)f(\theta_{\text{perm}})(W_{\text{perm}}^N(h) - U(h)) \right. \\ & \quad \left. + \pi_u f(\theta_j)(W_j^N(h - \Delta_h) - U(h - \Delta_h)) \right\} \end{aligned} \quad (17)$$

The wage equation (17) looks complicated. However, the basic intuition of the Nash Bargaining solution holds in this wage equation. The wage is the weighted average of the workers' outside options and firm profits. A worker's outside option is the unemployment income, b , plus the expected value of finding another job if the worker becomes unemployed. Note that there is a possibility that the workers' skill will drop to a level below h^* during the next period that they become unemployed. If their skills drops to a level below h^* , they will search in the temporary market. Thus, j in the last line of equation 17 could be either "perm" or "temp," depending on the workers' skill level. Firms and workers also take into consideration the value of being unemployed in the case of a skill upgrade if the match does not dissolve and the value of being unemployed in the case of a skill deterioration if the match is destroyed, both of which is captured in the equation's second line (17). Additionally, new workers are willing to accept lower initial wage if they can start working at firms and become existing workers next period. After they become existing workers, they can use the firing costs as a credible threat against the firm to demand a higher wage.

¹⁵See Appendix 9.4 for the derivation.

For existing permanent workers, the wages are given by:

$$\begin{aligned}
w_{\text{perm}}^E = & (1 - \eta)b + \eta(y_{\text{perm}} + (1 - \eta\beta(1 - \nu)\gamma_X(1 - \delta_{\text{perm}}))\eta\phi) \\
& + (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (\pi_{\text{perm}} - \pi_u)U(h) + \pi_u U(h - \Delta_h) - \pi_{\text{perm}} U(h + \Delta_h) \right\} \\
& + (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (1 - \pi_u)f(\theta_{\text{perm}})(W_{\text{perm}}^N(h) - U(h)) \right. \\
& \quad \left. + \pi_u f(\theta_j)(W_j^N(h - \Delta_h) - U(h - \Delta_h)) \right\}
\end{aligned} \tag{18}$$

Note that because existing permanent workers now know that their employers have to pay firing costs if the matches dissolve, the workers can demand higher wages. In our wage equation for the existing permanent worker, this appears in the form of $\eta\phi$ in the first line of equation (18), which gives the additional wage premium for the existing permanent worker.

For temporary workers, the wages are as follows:

$$\begin{aligned}
w_{\text{temp}} = & (1 - \eta)b + \eta(y_{\text{temp}}) \\
& + (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (\pi_{\text{temp}} - \pi_u)U(h) + \pi_u U(h - \Delta_h) - \pi_{\text{temp}} U(h + \Delta_h) \right\} \\
& + (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (1 - \pi_u)f(\theta_{\text{temp}})(W_{\text{temp}}^N(h) - U(h)) \right. \\
& \quad \left. + \pi_u f(\theta_{\text{temp}})(W_{\text{temp}}^N(h - \Delta_h) - U(h - \Delta_h)) \right\}
\end{aligned} \tag{19}$$

Since no firing cost is incurred for temporary jobs, ϕ does not enter the wage equation for temporary workers.

5.10 Dynamics of Distribution

Finally, the unemployment rate for each skill-level evolves as follows:

$$\begin{aligned}
u'(h) = & (1 - \pi_u)(1 - f(\theta_j)) \cdot u(h) + \pi_u(1 - f(\theta_j)) \cdot u(h + \Delta_h) \\
& + (1 - \pi_j)\delta_j \cdot (1 - u(h)) + \pi_j\delta_j \cdot (1 - u(h - \Delta_h)),
\end{aligned} \tag{20}$$

where j in equation 20 could be “perm” or “temp” depending on the skill level, h .

6 Calibration

The model is calibrated to match data at the monthly frequency. Table 2 summarizes the externally calibrated parameters.

Table 2: Externally calibrated parameters

Variable	Description	Value / Source
R	Interest Rate	$1.03072^{1/12}$, real interest rate between 1988-1994
β	Discount rate	$1/R$
δ_{perm}	Separation rate for perm workers	0.0035
δ_{temp}	Separation rate for temp workers	0.0071
γ_X	Technological growth rate	$1.0365^{1/12}$, growth rate between 1985-1994
θ_{perm}	Labor market tightness in perm sector	1.0, normalization
θ_{temp}	Labor market tightness in temp sector	2.24, Miyamoto (2016)
ψ	Matching elasticity	0.6, Miyamoto (2011)
η	Nash bargaining power for workers	0.6, see text
ν	Retirement probability	0.00208, see text
$\underline{h}$	min skill level	0
$\bar{h}$	max skill level	5
Δ_h	Human capital increment	0.0336, 150 equispaced grid
σ_h	std. of initial skill dist.	0.494, see text

I set the interest rate $R = 1.03072^{1/12}$, which was the average real rate of the 1-month overnight unsecured call rate between 1988 and 1994. For the discount factor, I simply take the inverse of R . I set the steady state rate of technological growth, $\gamma_X = 1.03027^{1/12}$, which translates into an average annual growth rate of 3.027%, which, was the Japanese average real GDP growth rate between 1985 and 1994.

I calibrate the average monthly separation rate for permanent workers to be 0.0035, which implies that their average years of working for a firm is 14.95 years. Following Miyamoto (2016), I calculate that the ratio of the separation rate for permanent workers to temporary workers is 0.49, as indicated in the Survey on Employment Trends conducted by the Ministry of Health, Labor and Welfare. I assume that matching efficiency parameters, $\bar{m}_j$ are different across sectors and that the elasticity parameters, ψ also are the same. Lin and Miyamoto (2012) estimated that the elasticity, ψ , for the Japanese labor market is 0.6. This value lies

in the plausible range of 0.5-0.7 that [Petrongolo and Pissarides \(2001\)](#) report. I set $\eta = 0.6$ according to the Hosios condition for efficiency.

For the benchmark calibration, I target the average monthly job finding rate of 0.155. This approximately corresponds to the hp-filtered job finding rate in Japan in 1990 as reported by [Miyamoto \(2011\)](#) and [Lin and Miyamoto \(2012\)](#). The Report on Employment Service conducted by the Ministry of Health, Labour and Welfare reports the job openings to application ratios of both permanent and temporary workers, which reflect labor market tightness. Based on the data and the estimates used by [Miyamoto \(2016\)](#), I target the ratio of labor market tightness for temporary to permanent $\theta_{\text{temp}}/\theta_{\text{perm}} = 2.24$. I normalize $\theta_{\text{perm}} = 1$. Each worker has an average working life of 40 years, which implies $\nu = 0.00208$. The maximum and minimum values of human capital $\underline{h}$ and $\bar{h}$ are chosen so that large masses of the skilled workers do not accumulate at the endpoints of the skill distribution. I set $\Delta_h = 0.0336$, using a grid that has 150 equispaced points.

Workers in my model are born with an initial skill level. I assume that the initial skill is drawn from a log-Normal distribution $\sim \ln N(0, \sigma_h^2)$, and I estimate σ_h by first running the following regression using the Japanese Household Panel Survey:

$$\log(\text{monthly income}_{i,t}) = \alpha + Z_{i,t}\beta + \eta_i + \epsilon_{i,t} \quad (21)$$

where $Z_{i,t}$ includes job-tenure (cubic function), work experience (cubic function), employment type (permanent, temporary), and fixed effects for city-size, year, and industry. Then, σ_h is the estimated permanent deviation of η_i in equation 21, which equals 0.494.

Table 4 lists the internally calibrated parameters. I estimate the following parameters by a simulated method of moments: ϕ , b , $\bar{m}_{\text{perm}}$, $\bar{m}_{\text{temp}}$, $\bar{A}$, π_{perm} , π_{temp} , and π_u . These parameters are estimated so that the model-produced moments match the data moments below. There are as many parameters as there are targeted moments. I begin by targeting the share of temporary workers to be 9.0%, which was the average share of temporary workers for Japanese men during the years 1990 to 1996. By focusing on the male population, I avoid the effects that arise when there is greater female participation in the labor force. Following [Miyamoto \(2016\)](#), I calculate from the Survey on Employment Trends conducted by Ministry of Health, Labor and Welfare that the ratio of job-finding rate of permanent workers to that of temporary workers is 0.45 . I also target the wage distribution of the p90/p50 ratio

to be 1.88, which, as estimated by [Lise et al. \(2014\)](#), were the average Japanese p90/p50 ratios prior to 1990. Using the estimated difference in return to working at the same firm between permanent workers and temporary workers provided in Section 3.1, I target the average difference in return on working between the two types of jobs to be 1%.¹⁶ Finally, estimating from the KHPS and the JHPS datasets, I target the average wage ratio between permanent workers and temporary workers to be 1.681. The list of data moments is displayed in Table 3.

Table 3: Targeted moments

Moment	Data (Target)	Model Output
Job finding rate	0.155	0.1551
Ratio of job finding rate: perm/temp	0.450	0.4496
Unemployment income	0.6729 (40% of average wage)	0.6704
Share of temporary workers	0.0900	0.0908
Wage distribution, p90/p50	1.879	1.8851
Wage distribution, p50/p10	1.706	1.7497
Average wage ratio: perm/temp	1.6808	1.3784
Average return on job tenure for perm workers	1.0000%	1.0536%

The estimated values of the parameters are shown in Table 4. Since the average wage of a permanent worker in my model is 15.619, the firing cost is estimated to be approximately equal to nine months of wage compensation.¹⁷ Following [Esteban-Pretel et al. \(2011\)](#) and [Miyamoto \(2016\)](#), I target the unemployment income to be about 40% of the average monthly wage of all workers, which is 0.6729. The estimated value of b is 0.6704. The estimated value of the matching efficiency in the permanent market and the temporary market implies that job finding rates in these sectors are 0.1431 and 0.2305, respectively. These

¹⁶The common wage growth component observed in Section 3.1 is captured in γ_X . The only margin that matters in my model is the *difference* between the rate of wage growth between permanent workers and temporary workers.

¹⁷[Kramarz and Michaud \(2010\)](#) estimated that in France the termination of the contract of a permanent job is 16% of the annual wage for an individual layoff and 50% of the annual wage for a collective layoff. Given that about four out of six layoffs are individual layoffs, the average cost is 20% of the annual wage. [Bentolila et al. \(2012\)](#) found that firing costs in Spain are about 20% higher than in France. [Alonso-Borrego et al. \(2005\)](#) estimated the firing cost are 51% of annual wages.

Table 4: Internally calibrated parameters

Variable	Description	Value
ϕ	Firing cost	15.619
b	Unemployment income	0.6704
$\bar{m}_{\text{perm}}$	Matching efficiency in permanent sector	0.1431
$\bar{m}_{\text{temp}}$	Matching efficiency in temporary sector	0.2305
$\bar{A}$	Productivity of temporary worker	1.2625
π_{perm}	Probability of skill upgrade for perm sector	0.0382
π_{temp}	Probability of skill upgrade for temp sector	0.0104
π_{u}	Probability of skill deterioration for unemployed	0.0209

numbers are targeted so that the job finding rate for the overall economy also matches the average monthly job-finding rate in Japan prior to 1990, which was approximately 0.155. The productivity of temporary workers, $\bar{A}$, is estimated to be 1.2625. Each month, the probabilities of skill upgrades for permanent jobs and temporary jobs are estimated to be 3.82% and 1.04%, respectively. If workers are unemployed, each month their skills deteriorate with the probability of 2.09%.

7 Results

To determine how slow growth affects the dual labor market, I now set the annual rate of technological growth to 0.6087%. This is roughly equivalent to the Japanese’s experience of economic growth during the lost decade(s). For the benchmark result, I also decrease the market interest rate so that it equals the average real interest in Japan between 1996 and 2014. Decreasing the real interest rate in my model is important because as the interest rate goes down, the future incomes are discounted at a lower rate, which reduces the effect of slow growth on the labor market. I take this into account when I conduct the experiment by decreasing the real interest rate to a level that we observe in the data. Table 5 summarizes the quantitative result.

In Table 5, column (1) presents the model moments when the rate of technological progress is 3.65% annually; and column (2) displays the model moments when the annual

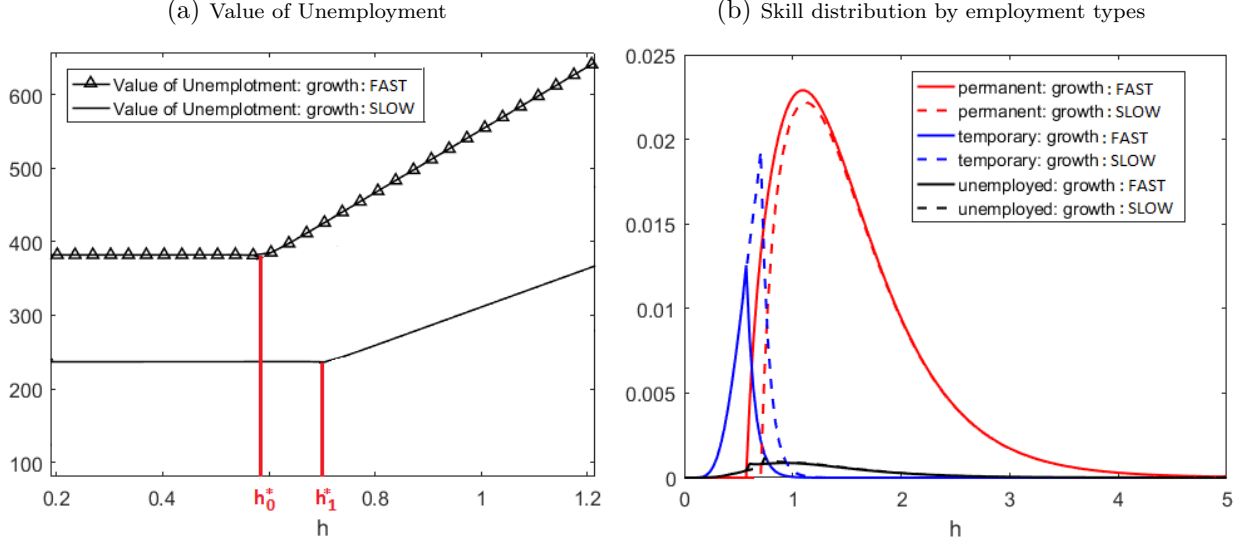
Table 5: Moments comparison under two regimes

Technological growth rate	(1) $\gamma_X = 1.0365^{1/12}$	(2) $\gamma_X = 1.006087^{1/12}$
Job finding rate	0.1551	0.1626
Job separation rate	0.0038	0.0041
Share of temporary worker	0.0908	0.1806
Average wage ratio: perm/temporary	1.3784	1.4363
Unemployment rate	3.67%	3.69%
Annual rate of interest	3.027%	0.414%
Market tightness in perm. sector	1.000	0.9268
Market tightness in temp. sector	2.240	2.0843
Job-finding rate in perm. sector	0.1431	0.1388
Job-finding rate in temp. sector	0.3183	0.3092

rate of technological progress is 0.61%. The share of temporary workers almost doubled from 9.08% to 18.06%. The average share of temporary workers for men in Japan increased from 9% between 1990 and 1996 to 20.42% between 2010 and 2015. The model attributes most of this increase to slow growth.

Figure 3a demonstrates the choices that unemployed workers face when deciding in which market to search for jobs. Figure 3a plots the values of unemployment for workers who have different skills for the fast growth steady state and for the slow growth steady state. The black line with triangles indicates the value of unemployment when the annual rate of economic growth is fast. The red line that is extending from the point, h_0^* , shows the original cutoff level or skill. The unemployed workers whose skill level is below h_0^* optimally search for temporary jobs, whereas the unemployed workers whose skill level is above h_0^* search in the permanent market. The value of unemployment is flat when workers are searching for temporary jobs because regardless of the workers' ability, they face the same job-finding probability and use the same production technology once they are employed. In contrast, the value of unemployment increases with skill level when workers are searching for permanent jobs because they can fully utilize their skills at the time of production. The black solid line indicates the value of unemployment in the 1% growth steady state. As the rate of economic growth slows down, the value of unemployment goes down. In this steady state, the cutoff

Figure 3: Baseline Calibration



level, h_1^* , shifts to the right, which induces a larger fraction of workers to search in the temporary market. The cutoff level moves to the right because the decrease in the value of unemployment in the permanent market is more significant than the decrease of the value of unemployment in the temporary market. The reduction is larger for the permanent market because the present value of wage growth decreases as all future incomes are discounted at a higher rate. Considering this, the unemployed workers whose skill level is within the range of (h_0^*, h_1^*) will switch from searching for jobs in the permanent market to searching for jobs in the temporary market. Thus, the share of temporary workers in the economy increases.

The market tightness in both sectors decreased, which indicates that firms post fewer vacancies per job-seeker when the economy is in the slow growth steady state. The decrease in market tightness leads to a low job-finding rate. Reinforcing the points made in the previous paragraph, Figure 3b clearly demonstrates that in the slower growth steady-state, the left tail of the skill distribution for workers in the permanent market shifted to the right. As the average skill levels of the permanent labor market increases, the model also predicts that the wage ratio between permanent and temporary employment increases in the slow growth steady-state.

Somewhat counter-intuitively, for the economy as a whole, the job-finding rate in the slower growth steady state increased slightly to 0.1626; in the faster growth steady-state

it was 0.1551. This is at odd with the observation made in previous paragraph where the job-finding rate within each market decreased. The improvement of the job-finding rate for overall market reflects the changes in the composition of the labor market, meaning that it is due to the large increase in the share of temporary workers in the economy. The job-finding rate is higher for temporary employment, and, thus, increasing the share of temporary workers improves the matching efficiency for the labor market overall. Likewise, the separation rate increases for the economy as a whole because a larger fraction of workers are now employed in the temporary jobs whose the job separation rate is higher.

Finally, the unemployment rate of the aggregate economy will increase in the slow growth steady state. This is consistent with the Japanese experience, wherein the unemployment rate has increased significantly during the last two decades. In this model, unemployment goes down because a greater number of workers search in a temporary market whose matching efficiency is higher.

7.1 No Endogenous Skill Accumulation

To highlight the importance of endogenous skill accumulation in my model, I lower the steady state growth rate in an economy from 4% to 1%, but I impose zero human capital accumulation (i.e. $\pi_{\text{perm}} = \pi_{\text{temp}} = \pi_u = 0$). In this environment workers are born with randomly realized skills, and these skills do not change for the rest of their careers. I recalibrate the model to match the moments listed in Appendix 9.5.

Table 6 summarizes the results of this experiment. The most striking result is that the share of temporary workers does not increase at all. Figure 4b shows that the change in the shape of the workers' skill distribution is very limited even after the steady-state growth rate has decreased to 1%. When human skills are not allowed to evolve, the rate of decrease in the value of searching in the permanent and of searching in the temporary sectors is symmetrical. As Figure 4a reveals, although the value of unemployment in the 4% growth steady state (black line with triangular) decreases when the economic growth becomes 1% (black solid line), the value of unemployment in the permanent market and the value of unemployment in the temporary market both go down by the same margin, which leaves the cutoff level h^* unaltered. In this case, workers will continue to compare the value of searching in the permanent market against the value of searching in the temporary market in the manner

Table 6: No Human Capital Accumulation
Moments comparison under two regimes

	(1)	(2)
Technological growth rate	$\gamma_X = 1.04^{1/12}$	$\gamma_X = 1.01^{1/12}$
Job finding rate	0.1585	0.1580
Ratio of job finding rate: perm/temporary	0.4496	0.4493
Job separation rate	0.0033	0.0033
Share of temporary worker	0.0937	0.0937
Unemployment rate	3.28%	3.29%
Annual rate of interest	1.040	1.0279
Job-finding rate in permanent sector	0.1461	0.1456
Job-finding rate in temporary sector	0.3249	0.3240

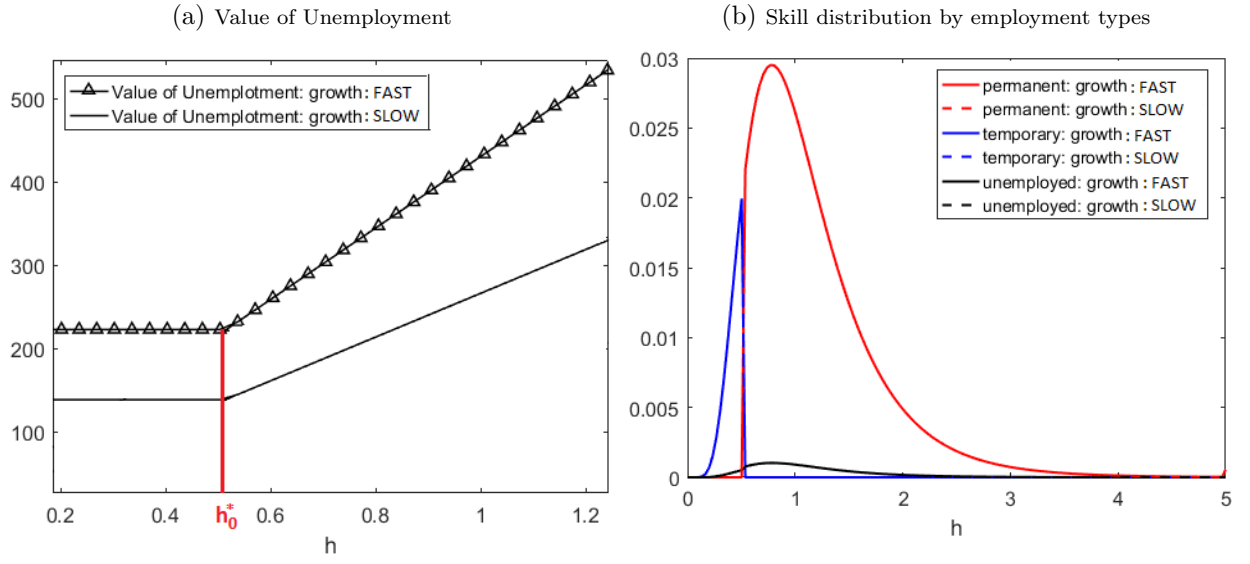
they did before the change occurred.

In this environment, the job-finding rate, the job separation rate, and the unemployment rate are not affected by the slowdown of economic growth as well. This section confirms that on-the-job training plays a key role in relationship between slow growth and labor market composition. The only source of wage growth in this model is skill accumulation. Thus, the steeper the rate of wage growth for permanent workers, the larger the impact on labor market composition of changes in the growth rate.

8 Concluding Remarks

This paper investigates how the rate of economic growth affects the composition of the labor market. I build a search and matching model of a dual labor market that is inhabited by heterogeneous workers. I find that the rate of workers' skill accumulation on-the-job significantly affects the manner in which slow growth impacts labor market composition. Specifically, the higher the rate of skill accumulation for permanent workers, the greater the impact of slow growth on labor market composition. My model attributes the majority of the observed increase in the share of Japan's temporary workers to that country's slow growth. When a faster rate of skill accumulation is combined with slower economic growth,

Figure 4: No Human Capital Accumulation



the labor market condition can change dramatically.

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9 Appendix

9.1 Data Description

Throughout the analysis, I used the Japanese Household Panel Survey (JHPS) and the Keio Household Panel Survey (KHPS). The original datasets include approximately 4000 households between 2004 to 2008 and 8000 households between 2009 and 2014. I focus on the income of sub-sample of the household-heads who were recorded at least 4 times. This criterion leaves us with about 2000 samples each year from 2004 to 2008 and about 3200 samples from 2009 to 2013.

The survey records the labor income of household-heads under the following four categories: (i) monthly wages, (ii) weekly wages, (iii) hourly wages, and (iv) annual bonuses. I construct an estimated hourly wages based on all four measures of the wages.

The monthly wages and annual bonuses are converted into hourly wages by the following procedure:

$$\text{hourly wages} = \frac{\text{monthly wages} + \text{annual bonuses} / 12}{(\text{average days worked per month} / 4.381 \text{ weeks}) \times \text{hours of work per week}}$$

The denominator is the hours worked per month, which is estimated using the reported average days worked per month, the average number of weeks in a month (4.381 weeks), and the reported hours of work per week.

The weekly wages are converted into hourly wages by the following procedure:

$$\text{hourly wages} = \frac{\text{weekly wages}}{\text{hours of work per week}}$$

A part of the job-tenure is also estimated for the analysis. The survey asks respondents the year in which they started the current employment. From this figure, I calculate the job tenure. However, this question is available only when the respondents enter the survey for the first time. For the following years, I add a year to the job tenure if the respondents reported that they stayed at the same workplace from previous year. Otherwise, I set the job tenure to 0.

The work experience is constructed in a similar manner. The number of years working

is reported only when the respondents are recorded in the survey for the first time. From this, I add a year to the work experience if the respondents reported that they worked in the following years. Otherwise, the work experience in each survey year remains the same as the previous year.

9.2 Stationary Inducing Transformation

This section shows the derivation of the stationary inducing transformation and that the value functions are homogeneous of degree one in X . Let variables in *tilde* indicate the variables before the stationary-inducing transformation. Without loss of generality, I assume no human capital accumulation, $\pi_j = 0$ for $j \in \{\text{perm}, \text{temp}, \text{u}\}$, and no outside wages for permanent workers.¹⁸ The system of value functions before conducting the stationary-inducing transformation is:

$$\tilde{U}(h) = \tilde{b} + \beta(1 - \nu) \max \mathbb{E} \left[\tilde{U}'_{\text{perm}}(h), \tilde{U}'_{\text{temp}}(h) \right] \quad (22)$$

$$\tilde{U}_{\text{perm}}(h) = f(\theta_{\text{perm}})(\tilde{W}_{\text{perm}}(h) - \tilde{U}(h)) + \tilde{U}(h) \quad (23)$$

$$\tilde{U}_{\text{temp}}(h) = f(\theta_{\text{temp}})(\tilde{W}_{\text{temp}}(h) - \tilde{U}(h)) + \tilde{U}(h) \quad (24)$$

$$\tilde{J}_{\text{perm}}(h) = h \times X - \tilde{w}_{\text{perm}}(h) + \beta(1 - \nu) \mathbb{E} \left[(1 - \delta_{\text{perm}}) \tilde{J}'_{\text{perm}}(h) - \delta_{\text{perm}} \tilde{\phi}' \right] \quad (25)$$

$$\tilde{J}_{\text{temp}}(h) = A \times X - \tilde{w}_{\text{temp}}(h) + \beta(1 - \nu) \mathbb{E} \left[(1 - \delta_{\text{temp}}) \tilde{J}'_{\text{perm}}(h) \right] \quad (26)$$

$$\kappa_{\text{perm}} \tilde{\text{perm}} = \beta(1 - \nu) \mathbb{E} \left[q(\theta_{\text{perm}}) \int_h \tilde{J}'_{\text{perm}}(h) d\mu_{\text{perm}}^h \right] \quad (27)$$

$$\kappa_{\text{temp}} \tilde{\text{temp}} = \beta(1 - \nu) \mathbb{E} \left[q(\theta_{\text{temp}}) \int_h \tilde{J}'_{\text{temp}}(h) d\mu_{\text{temp}}^h \right] \quad (28)$$

$$\tilde{w}_{\text{perm}} = (1 - \eta) \tilde{b} + \eta \left(h \times X - \beta(1 - \nu)(1 - \delta_{\text{perm}}) \tilde{\phi}' \right) \quad (29)$$

$$+ (1 - \eta) \beta(1 - \nu) \left\{ f(\theta_{\text{perm}})(\tilde{W}'_{\text{perm}}(h) - \tilde{U}'(h)) \right\}$$

$$\tilde{w}_{\text{temp}} = (1 - \eta) \tilde{b} + \eta (A \times X) \quad (30)$$

$$+ (1 - \eta) \beta(1 - \nu) \left\{ f(\theta_{\text{temp}})(\tilde{W}'_{\text{temp}}(h) - \tilde{U}'(h)) \right\}$$

¹⁸Under this assumption, the permanent firms incur firing costs as soon as firms and workers meet at the initial period of employment.

To derive a system of equations that are on the balanced growth, I divide all variables by X . I assume that X exhibits a steady-state rate of growth denoted by γ_X , $\gamma_X = X'/X$. Furthermore, I also assume that all exogenous variables grow at the constant rate of γ_X : $b = \tilde{b}/X$, $\phi = \tilde{\phi}/X$, and $\kappa_j = \tilde{\kappa}_j/X$. Under these assumptions, the system of equations (22)-(30) can be written as follows:

$$U(h) = b + \beta(1 - \nu)\gamma_X \max \mathbb{E} [U_{\text{perm}}(h), U_{\text{temp}}(h)] \quad (31)$$

$$U_{\text{perm}}(h) = f(\theta_{\text{perm}})(W_{\text{perm}}(h) - U(h)) + U(h) \quad (32)$$

$$U_{\text{temp}}(h) = f(\theta_{\text{temp}})(W_{\text{temp}}(h) - U(h)) + U(h) \quad (33)$$

$$J_{\text{perm}}(h) = h - w_{\text{perm}}(h) + \beta(1 - \nu)\gamma_X \mathbb{E} \left[(1 - \delta_{\text{perm}})J_{\text{perm}}(h) - \delta_{\text{perm}}\phi \right] \quad (34)$$

$$J_{\text{temp}}(h) = A - w_{\text{temp}}(h) + \beta(1 - \nu)\gamma_X \mathbb{E} \left[(1 - \delta_{\text{temp}})J_{\text{perm}}(h) \right] \quad (35)$$

$$\kappa_{\text{perm}} = \beta(1 - \nu)\gamma_X \mathbb{E} \left[q(\theta_{\text{perm}}) \int_h J_{\text{perm}}(h) d\mu_{\text{perm}}^h \right] \quad (36)$$

$$\kappa_{\text{temp}} = \beta(1 - \nu)\gamma_X \mathbb{E} \left[q(\theta_{\text{temp}}) \int_h J_{\text{temp}}(h) d\mu_{\text{temp}}^h \right] \quad (37)$$

$$w_{\text{perm}} = (1 - \eta)b + \eta(h - \beta(1 - \nu)(1 - \delta_{\text{perm}})\gamma_X\phi) \quad (38)$$

$$+ (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ f(\theta_{\text{perm}})(W_{\text{perm}}(h) - U(h)) \right\}$$

$$w_{\text{temp}} = (1 - \eta)b + \eta(A) \quad (39)$$

$$+ (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ f(\theta_{\text{temp}})(W_{\text{temp}}(h) - U(h)) \right\}$$

Thus, the system of equations are homogeneous of degree one in X .

9.3 Wage Determination

As explained in Section 5.9, a wage is negotiated between a worker and a firm according to the Nash Bargaining. The two parties split the match surplus as follows: For a new permanent worker

$$(1 - \eta)(W_{\text{perm}}^N - U_{\text{perm}}) = \eta J_{\text{perm}}^N \quad (40)$$

For an existing permanent worker

$$(1 - \eta)(W_{\text{perm}}^E - U_{\text{perm}}) = \eta(J_{\text{perm}}^E + \phi) \quad (41)$$

And for a temporary worker

$$(1 - \eta)(W_{\text{temp}} - U_{\text{temp}}) = \eta J_{\text{temp}} \quad (42)$$

I derive the wage for a new permanent worker here. To derive the wage equation, I first write out the value functions in the equation 40 using the value functions defined in sections 5.5, 5.6, and 5.7:

$$\begin{aligned} & (1 - \eta) \left(w_{\text{perm}}^N + \beta(1 - \nu)\gamma_X \left\{ (1 - \pi_{\text{perm}})(1 - \delta_{\text{perm}}) ((W_{\text{perm}}^E(h, z) - U(h)) + U(h)) \right. \right. \\ & \quad \left. \left. + \pi_{\text{perm}}(1 - \delta_{\text{perm}}) ((W_{\text{perm}}^E(h + \Delta_h, z) - U(h + \Delta_h)) + U(h + \Delta_h)) \right\} \right. \\ & \quad \left. - b - \beta(1 - \nu)\gamma_X \left\{ (1 - \pi_u) (f(\theta_{\text{perm}})(W_{\text{perm}}^N(h, z) - U(h)) + U(h)) \right. \right. \\ & \quad \left. \left. + \pi_u (f(\theta_{\text{perm}})(W_{\text{perm}}^N(h - \Delta_h, z) - U(h - \Delta_h)) + U(h - \Delta_h)) \right\} \right) \\ & = \eta \left(y_{\text{perm}}^N - w_{\text{perm}}^N \right. \\ & \quad \left. + \beta(1 - \nu)\gamma_X \left\{ (1 - \pi_{\text{perm}})((1 - \delta_{\text{perm}})J_{\text{perm}}^E(h, z) - \phi\delta_{\text{perm}}) \right. \right. \\ & \quad \left. \left. + \pi_{\text{perm}}((1 - \delta_{\text{perm}})J_{\text{perm}}^E(h + \Delta_h, z) - \phi\delta_{\text{perm}}) \right\} \right) \end{aligned}$$

Using the expression in equation 41, I can simplify the expression above as follows:

$$\begin{aligned}
&\Rightarrow \\
&w_{\text{perm}}^N - (1 - \eta)b - \eta(y_{\text{perm}}^N - Rf'(k)) \\
&+ \beta(1 - \nu)\gamma_X \left\{ (1 - \pi_{\text{perm}})(1 - \delta_{\text{perm}}) \underbrace{\left((1 - \eta) (W_{\text{perm}}^E(h, z) - U(h)) - \eta J_{\text{perm}}^E(h, z) \right)}_{\eta\phi} \right. \\
&+ \pi_{\text{perm}}(1 - \delta_{\text{perm}}) \underbrace{\left((1 - \eta) (W_{\text{perm}}^E(h + \Delta_h, z) - U(h + \Delta_h)) - \eta J_{\text{perm}}^E(h + \Delta_h, z) \right)}_{\eta\phi} \\
&+ (1 - \eta) \left((1 - \pi_{\text{perm}})U_{\text{perm}}(h) + \pi_{\text{perm}}U_{\text{perm}}(h + \Delta_h) - (1 - \pi_u)U_{\text{perm}}(h) - \pi_u U_{\text{perm}}(h - \Delta_h) \right) \\
&\left. - (1 - \pi_u)f(\theta_{\text{perm}})\eta J_{\text{perm}}^N(h, z) - \pi_u f(\theta_{\text{perm}})\eta J_{\text{perm}}^N(h - \Delta_h, z) \right\} = 0
\end{aligned}$$

Finally, the wage equation for a new permanent worker can be written as follows:

$$\begin{aligned}
w_{\text{perm}}^N &= (1 - \eta)b + \eta(y_{\text{perm}} - \beta(1 - \nu)\gamma_X(1 - \delta_{\text{perm}})\phi) \\
&+ (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (\pi_{\text{perm}} - \pi_u)U(h) + \pi_u U(h - \Delta_h) - \pi_{\text{perm}}U(h + \Delta_h) \right\} \\
&+ (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (1 - \pi_u)f(\theta_{\text{perm}})(W_{\text{perm}}^N(h) - U(h)) \right. \\
&\quad \left. + \pi_u f(\theta_j)(W_j^N(h - \Delta_h) - U(h - \Delta_h)) \right\}
\end{aligned}$$

One can follow the same steps to derive the wage equation for an existing permanent

worker:

$$\begin{aligned}
w_{\text{perm}}^E = & (1 - \eta)b + \eta(y_{\text{perm}} + (1 - \eta\beta(1 - \nu)\gamma_X(1 - \delta_{\text{perm}}))\eta\phi) \\
& + (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (\pi_{\text{perm}} - \pi_u)U(h) + \pi_u U(h - \Delta_h) - \pi_{\text{perm}} U(h + \Delta_h) \right\} \\
& + (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (1 - \pi_u)f(\theta_{\text{perm}})(W_{\text{perm}}^N(h) - U(h)) \right. \\
& \quad \left. + \pi_u f(\theta_j)(W_j^N(h - \Delta_h) - U(h - \Delta_h)) \right\}
\end{aligned}$$

Likewise, the wage equation for a temporary worker can be derived as follows:

$$\begin{aligned}
w_{\text{temp}} = & (1 - \eta)b + \eta(y_{\text{temp}}) \\
& + (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (\pi_{\text{temp}} - \pi_u)U(h) + \pi_u U(h - \Delta_h) - \pi_{\text{temp}} U(h + \Delta_h) \right\} \\
& + (1 - \eta)\beta(1 - \nu)\gamma_X \left\{ (1 - \pi_u)f(\theta_{\text{temp}})(W_{\text{temp}}^N(h) - U(h)) \right. \\
& \quad \left. + \pi_u f(\theta_{\text{temp}})(W_{\text{temp}}^N(h - \Delta_h) - U(h - \Delta_h)) \right\}
\end{aligned}$$

9.4 Computation

The computation algorithm for the model with exogenous separation is explained in this section. There is only one state variable, which is workers' skill-level, h . Because workers are heterogeneous only in their skills, workers with the same skill all search in the same market in the steady state.

The computation algorithm is explained as follows:

1. Guess, for all h , the market in which the workers with skill level h search for a job
2. Guess $U(h)$ for each skill level, h .
3. Solve for J and w using Equations 9, 10, 11, 17, 18, and 19.

4. Solve for θ and the steady-state skill distributions of workers using Equations 12, 13, and 20.
5. Solve for W using Equations 6, 7, and 8.
6. Verify that guess of $U(h)$ satisfies Equations 4 and 5 for all skill levels, h ,.
 - If not, update the guess, and return to step 3. Otherwise, proceed to step 7.
7. Verify that initial guess of market that workers with skill, h , search satisfies Equation 3
 - If not, update the guess, and return to step 1. Repeat until convergence.

9.5 Calibration strategy for No Human Capital Accumulation Economy

Table 7 summarizes the target moments for calibrating the model in an economy that prohibits endogenous skill accumulation. The task of the procedure is to find values of the five parameters, ϕ , b , $\bar{m}_{\text{perm}}$, $\bar{m}_{\text{temp}}$, and A in order to match five moments. The estimated parameter values are listed in Table 8.

Table 7: Targets and Model Moments

Moment	Data (Target)	Model Output
Job finding rate	0.155	0.1585
Ratio of job finding rate: perm/temp	0.450	0.4496
Job separation rate	0.0035	0.0033
Share of temporary worker	0.080	0.0937
Average wage ratio: perm/tem	1.681	2.2272

9.6 other possible explanation in Japan

Industry composition Legislative change Uncertainty

Table 8: Internally calibrated parameters

Variable	Description	Value
ϕ	Firing cost	0.0375
b	Unemployment income	0.4500
$\bar{m}_{\text{perm}}$	Matching efficiency in the permanent sector	0.1461
$\bar{m}_{\text{temp}}$	Matching efficiency in the temporary sector	0.2353
A	Productivity of temporary worker	0.5335